

## ON A CLASS OF RANK TESTS FOR LACK OF FIT

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### SUMMARY

A class of aligned rank order tests for lack of fit of the least squares plane

$$X^{(1)} = \beta_1 + \sum_{r=2}^c \beta_r x^{(r)}$$

is proposed. The asymptotic distribution under the null-hypothesis (of linear fit) and contiguous alternatives is found. Asymptotic relative efficiency is investigated and the practical implementation is discussed.

### 1. Introduction

In recent literature on rank tests much attention has been paid to aligned tests for regression problems. In Sen (1969), Adichie (1974), Sen (1972) and Adichie (1975) tests for parallelism of regression lines and for the identity of intercepts and/or slopes were considered. Sen and Puri (1977) discussed a class of aligned rank tests for composite hypotheses in multivariate general linear models, which among other applications included a multivariate extension of Sen (1969). Adichie (1978) proposed a class of rank tests and a class of signed rank tests for testing of subhypotheses in the univariate general linear model, and illustrated the use of the latter class when testing a bivariate linear relationship against a bivariate quadratic relationship. Olshen (1967) used a different approach when he proposed sign and Wilcoxon tests for this problem. The exact distributions of the test statistics of his tests are known but these statistics are not uniquely determined, since they are calculated from certain randomly chosen observations.

In the present study a class of aligned rank tests for lack of fit of a least squares plane is proposed. The asymptotic distribution of the test statistic is found under the null hypothesis of linear fit and contiguous alternatives and the asymptotic relative efficiency (when compared to the classical least squares competitor) is determined. Finally, the practical implementation of the proposed procedure is briefly discussed.

### 2. Notation and definition of the test statistic

The problem we consider is testing lack of fit of the least squares plane

$$X^{(1)} = \beta_1 + \sum_{r=2}^c \beta_r X^{(r)}$$

to the  $n$  observations  $(X_{i\alpha}^{(1)}, x_i^{(2)}, \dots, x_i^{(c)})$ ,  $\alpha = 1, 2, \dots, n_i$ ;  $i = 1, 2, \dots, k$  with  $\sum_{i=1}^k n_i = n$  and  $c < k$ .

More formally, testing of

$$(1) \quad H_0 : X_{i\alpha}^{(1)} = \beta_1 + \sum_{r=2}^c \beta_r x_i^{(r)} + E_{i\alpha}$$

against a general class of alternatives, contradicting the null hypothesis, is considered. It will be assumed that the  $E_{i\alpha}$  ( $\alpha = 1, 2, \dots, n_i$ ;  $i = 1, 2, \dots, k$ ) are independent identically distributed variables with continuous c.d.f. (cumulative distribution function)  $F$ , zero expectation and unknown variance  $\sigma^2$ . The  $\beta$ 's are unknown constants and the  $x_i^{(r)}$  known regression constants.

Define

$$(2) \quad \bar{X}_i^{(1)} = n_i^{-1} \sum_{\alpha=1}^{n_i} X_{i\alpha}^{(1)}; \bar{X}^{(1)} = n^{-1} \sum_{i=1}^k n_i \bar{X}_i^{(1)}$$

$$\bar{x}^{(r)} = n^{-1} \sum_{i=1}^k n_i x_i^{(r)}, r = 2, 3, \dots, c$$

$$(3) \quad S_{1r} = \sum_{i=1}^k n_i (x_i^{(r)} - \bar{x}^{(r)}) (\bar{X}_i^{(1)} - \bar{X}^{(1)})$$

$$s_{rt} = \sum_{i=1}^k n_i (x_i^{(r)} - \bar{x}^{(r)}) (x_i^{(t)} - \bar{x}^{(t)})$$

$$r, t = 2, 3, \dots, c.$$

Let

$$S = \begin{bmatrix} S_{11} & S_{12} & \dots & S_{1c} \\ S_{21} & s_{22} & \dots & s_{2c} \\ \vdots & \vdots & \ddots & \vdots \\ S_{c1} & s_{c2} & \dots & s_{cc} \end{bmatrix}$$

and denote by  $|S_{11}|$  the cofactor in  $|S|$  of  $S_{11}$ . Throughout the paper it will be assumed

that

$$(A) \quad \lim_{n \rightarrow \infty} \frac{n_i}{n} = \lambda_i, \text{ with } 0 < \lambda_0 \leq \lambda_i \leq 1 - \lambda_0 < 1 \text{ and } \lambda_0 \text{ some constant less than } \frac{1}{2};$$

$$(B) \quad |S_{11}| \neq 0; \\ \text{and for some sequence of non-zero constants } a_n:$$

$$(C) \quad s_{rt}/a_n = O(1);$$

$$(D) \quad \lim_{n \rightarrow \infty} a_n = \infty.$$

It will also be assumed that  $\alpha, \beta = 1, 2, \dots, n_i$  (or  $n_j$  or  $n_h$ ) and  $h, i, j = 1, 2, \dots, k$ . Further define

$$(4) \quad |W_i| = n_i / |S_{11}| \quad \begin{vmatrix} (\bar{x}_i^{(1)} - \bar{x}^{(1)}) & S_{12} & \dots & S_{1c} \\ (x_i^{(2)} - \bar{x}^{(2)}) & s_{22} & \dots & s_{2c} \\ \vdots & & & \\ (x_i^{(c)} - \bar{x}^{(c)}) & s_{c2} & \dots & s_{cc} \end{vmatrix}$$

and denote by  $|W_i^{(r)}|$  the cofactor in  $|W_i|$  of  $S_{1r}$ . Through routine matrix algebra it follows that

$$(5) \quad \sum_i |W_i^{(r)}| = 0$$

$$(6) \quad \sum_i (x_i^{(t)} - \bar{x}^{(t)}) |W_i^{(r)}| = -\delta_{rt}$$

with  $\delta_{rt}$  the Kronecker delta;

$$(7) \quad \sum_i n_i^{-1} |W_i^{(r)}| \cdot |W_i^{(t)}| = |S_{11 \cdot rt}| / |S_{11}|$$

with  $|S_{11 \cdot rt}|$  the cofactor in  $|S_{11}|$  of  $s_{rt}$ , where we assume that  $c \geq 3$ , and

$$(8) \quad \sum_i n_i (x_i^{(t)} - \bar{x}^{(t)}) |S_{11 \cdot rt}| / |S_{11}| = -|W_i^{(r)}|.$$

Now define

$$(9) \quad c_{ij} = \sqrt{n_i n_j}^{-1} \{ \delta_{ij} - n_j/n + \sum_r (x_i^{(r)} - \bar{x}^{(r)}) |W_j^{(r)}| \}$$

then it follows from the expressions above that

$$(10) \quad \sum_j n_j c_{ij} = 0$$

$$(11) \quad \sum_j n_j c_{ij} x_j^{(r)} = 0$$

$$(12) \quad \sum_h n_h c_{ih} c_{jh} = \delta_{ij} - \{ \sqrt{n_i n_j} / n + \sum_r \sum_t \sqrt{n_i n_j} (x_i^{(r)} - \bar{x}^{(r)}) (x_j^{(t)} - \bar{x}^{(t)}) |S_{11 \cdot rt}| / |S_{11}| \}.$$

Consider the estimators  $\hat{\beta}_{nr}$  of  $\beta_r$  ( $r = 2, 3, \dots, c$ ) which satisfy a regular condition to be given in the next section (see (20)) and define

$$(13) \quad \hat{X}_{j\alpha} = X_{j\alpha}^{(1)} - \sum_r \hat{\beta}_{nr} x_j^{(r)}$$

$$(14) \quad X_{j\alpha}^0 = X_{j\alpha}^{(1)} - \sum_r \beta_r x_j^{(r)}$$

$$(15) \quad X_{j\alpha}^* = X_{j\alpha}^{(1)} - \sum_r (\beta_r + b_r / \sqrt{a_n}) x_j^{(r)}$$

where  $|b_r| \leq M$ , a constant.

Let  $\hat{R}_{j\alpha}$ ,  $R_{j\alpha}^0$  and  $R_{j\alpha}^*$  ( $\alpha = 1, 2, \dots, n_j$ ;  $j = 1, 2, \dots, k$ ) be the ranks of the

$$\hat{X}_{j\alpha}'s, X_{j\alpha}^{0'}s \text{ and } X_{j\alpha}^{*'}s$$

respectively. Also, define

$$(16) \quad \hat{T}_{ni} = \sum_j \sum_{\alpha} c_{ij} \psi \{ \hat{R}_{j\alpha} / (n+1) \}$$

where  $\psi(u)$ ,  $0 < u < 1$ , is a continuous nondecreasing function. Similarly, define  $T_{ni}^0$  and  $T_{ni}^*$  through the  $X_{j\alpha}^{0'}$ 's and  $X_{j\alpha}^{*'}s$  respectively. Finally, define

$$(17) \quad A_{\psi}^2 = \int_0^1 \psi^2(u) du - \{ \int_0^1 \psi(u) du \}^2.$$

As test statistic for  $H_0$  we shall use

$$(18) \quad \hat{Q}_n = \sum_i \hat{T}_{ni}^2 / A_{\psi}^2.$$

### 3. Asymptotic joint distribution of the $\{\hat{T}_{ni}\}$

In this section the asymptotic joint distribution of the  $\{\hat{T}_{ni}\}$  is derived under  $H_0$  and a class of contiguous alternatives

(19)  $H_n : X_{i\alpha}^{(1)} = \beta_1 + \sum_r \beta_r X_i^{(r)} + E_{i\alpha} + \theta_i / \sqrt{a_n}$  where the  $\theta_i$  are arbitrary constants and  $a_n$  satisfies conditions C and D of the preceding section. It will be assumed that the estimators  $\hat{\beta}_{nr}$  satisfy the following condition under  $H_0$  and  $H_n$ :

$$(20) \quad \lim_{n \rightarrow \infty} P \{ |\sqrt{a_n} (\hat{\beta}_{nr} - \beta_r)| < A \} = 1$$

where  $A$  is a positive constant. The following lemma is similar to Lemma 2.2 of Adichie (1974).

*Lemma 1*

*If*

(i)  $\psi(u)$  possesses a bounded second derivative;

(ii)  $\sup_y [\psi'_y \{F(y)\}]$  is bounded where  $\psi'_y \{F(y)\} \equiv \frac{d}{dy} \psi \{F(y)\}$

(iii)  $(\max_{r,i} |x_i^{(r)}|) / (\max_{r,i} |x_i^{(r)} - \bar{x}^{(r)}|) = O(1)$

then  $E(\hat{T}_{ni} - T_{ni}^0)^2 \xrightarrow{n} 0$  in probability under  $H_0$  and  $H_n$ .

*Proof*

Without loss of generality we assume that  $\beta_r = 0$ ,  $r = 1, 2, \dots, c$ . From (15) and (20) we see that  $\hat{X}_{j\alpha}$  and  $X_{j\alpha}^*$  tend to  $X_{j\alpha}^0$  at the same rate, so that it will be sufficient to prove the lemma in terms of  $T_{ni}^*$  instead of  $\hat{T}_{ni}$ . We also note that conditions (A) and (C) imply that

$$(21) \quad (\max_{r,i} |x_i^{(r)} - \bar{x}^{(r)}|) \sqrt{n_i} / \sqrt{a_n} = O(1).$$

Following Hajek (1968) and Adichie (1974), define

$$(22) \quad Z_{ni}^0(X_{j\alpha}^0) = n^{-1} \sum_h n_h (c_{ih} - c_{ij}) \int_{-\infty}^{+\infty} \{u(x - X_{j\alpha}^0) - F_j^0(x)\} \psi' \{H^0(x)\} dF_h^0(x)$$

with  $u(x) = 1$  when  $x \geq 0$  and  $0$  when  $x < 0$  and where

$$H^0(x) = \sum_j (n_j/n) F_j^0(x) \text{ with } F_j^0(x) \text{ the c.d.f. of } X_{j\alpha}^0$$

$$= n^{-1} \sum_h n_h (c_{ih} - c_{ij}) [-\psi \{F_j^0(X_{j\alpha}^0)\} + K_{jh}^0 + Q_{jh}^0(X_{j\alpha}^0)]$$

$$\text{with } |Q_{jh}^0(X_{j\alpha}^0)| \leq L \max_{j, h; y} |F_j^0(y) - F_h^0(y)|,$$

$L$  and  $K_{jh}^0$  being constants independent of  $n$ .

Let

$$(23) \quad \mu_{ni}^0 = \sum_j n_j c_{ij} \int_{-\infty}^{+\infty} \psi \{H^0(y)\} dF_j^0(y)$$

and let  $Z_{ni}^*(X_{j\alpha}^*)$  and  $\mu_{ni}^*$  similarly be defined in terms of  $X_{j\alpha}^*$  and  $F_j^*(x)$ , the c.d.f. of  $X_{j\alpha}^*$

Since

$$\begin{aligned} E_n (T_{ni}^* - T_{ni}^0)^2 & \leq 4E_n [(T_{ni}^* - \sum_j \sum_\alpha Z_{ni}^*(X_{j\alpha}^*) - \mu_{ni}^*)^2 + (T_{ni}^0 - \sum_j \sum_\alpha Z_{ni}^0(X_{j\alpha}^0) - \mu_{ni}^0)^2 \\ & \quad + \{\sum_j \sum_\alpha \{Z_{ni}^*(X_{j\alpha}^*) - Z_{ni}^0(X_{j\alpha}^0)\}^2 + (\mu_{ni}^0 - \mu_{ni}^*)^2\}] \end{aligned}$$

we only need to prove that

$$(24) \quad E_n (T_{ni}^* - \sum_j \sum_\alpha Z_{ni}^*(X_{j\alpha}^*) - \mu_{ni}^*)^2 \rightarrow 0$$

$$(25) \quad E_n [\sum_j \sum_\alpha \{Z_{ni}^*(X_{j\alpha}^*) - Z_{ni}^0(X_{j\alpha}^0)\}^2] \rightarrow 0$$

and

$$(26) \quad (\mu_{ni}^0 - \mu_{ni}^*)^2 \rightarrow 0$$

where  $E_n$  denotes expectation under  $H_n$ .

Since  $|S_{11 \cdot r t}|/|S_{11}| = O(a_n^{-1})$ , we see from (21) that  $\sum_j n_j c_{ij}^2$  is bounded and (24) thus follows from (4.26) and (4.27) of Hajek (1968).

Similarly for  $E_n (T_{ni}^0 - \sum_j \sum_\alpha Z_{ni}^0(X_{j\alpha}^0) - \mu_{ni}^0)^2$ .

To prove (25) we note that  $Z_{ni}^*(X_{j\alpha}^*)$  and  $Z_{ni}^0(X_{j\alpha}^0)$  have zero expectation and

$$\{Z_{ni(j, \alpha)} \equiv Z_{ni}^*(X_{j\alpha}^*) - Z_{ni}^0(X_{j\alpha}^0), \alpha = 1, 2, \dots, n_j; j = 1, 2, \dots, k\}$$

is a set of independent variables.

Thus

$$\begin{aligned} (27) \quad E_n \{ \sum_j \sum_\alpha Z_{ni(j, \alpha)} \}^2 \\ = \sum_j \sum_\alpha \text{var} [n^{-1} \sum_h n_h (c_{ih} - c_{ij}) \{ \psi(F_j^0(X_{j\alpha}^0)) - \psi(F_j^*(X_{j\alpha}^*)) \\ + K_{jh}^* - K_{jh}^0 + Q_{jh}^*(X_{j\alpha}^*) - Q_{jh}^0(X_{j\alpha}^0) \}] \\ \leq \sum_j \sum_\alpha n^{-1} \sum_h n_h (c_{ih} - c_{ij})^2 E[\psi\{F_j^0(X_{j\alpha}^0)\} - \psi\{F_j^*(X_{j\alpha}^*)\} + Q_{jh}^*(X_{j\alpha}^*) - Q_{jh}^0(X_{j\alpha}^0)]^2. \end{aligned}$$

Now, since  $|\psi\{F_j^*(y)\} - \psi\{F_j^0(y)\}|$

$$\leq \text{const.} (\max_{r,j} |x_j^{(r)}| / \sqrt{a_n}) \sup_y |\psi'_y\{F(y)\}|$$

$\rightarrow 0$  from conditions (ii) and (iii) and (21)

and since

$$|Q_{jh}^*(y) - Q_{jh}^0(y)| = O(a_n^{-1/2}) \text{ under } H_n \text{ for all } y \text{ and also}$$

$$n^{-1} \sum_j n_j \sum_h n_h (c_{ih} - c_{ij})^2 = 2 \sum_j n_j c_{ij}^2, (25) \text{ follows.}$$

Finally, under  $H_n$  the following holds for the c.d.f. of  $X_{j\alpha}^0$ :

$$F_h^0(y) = F_j^0(y + \rho_{n;jh}) \text{ with } \rho_{n;jh} = (\theta_j - \theta_h) / \sqrt{a_n}.$$

On applying the multivariate mean value theorem, one finds

$$\begin{aligned}\mu_{ni}^0 &= \sum_j n_{j,c} \int_{-\infty}^{+\infty} \psi \{F_j^0(y)\} dF_j^0(y) \\ &+ \sum_j \sum_h n_{j,c} \rho_{n;jh} \int_{-\infty}^{+\infty} \left[ \frac{\partial}{\partial \rho_{n;jh}} \psi \{ \sum_h (n_h/n) F_j^0(y + \rho_{n;jh}) \} \right] \rho'_{n;jh} dF_j^0(y)\end{aligned}$$

where  $|\rho'_{n;jh}| \leq |\rho_{n;jh}|$

$$= \sum_j \sum_h n_{j,c} \rho_{n;jh} \int_{-\infty}^{+\infty} \frac{d}{dq} \psi(q) \frac{dq}{d\rho_{n;jh}} dF_j^0(y)$$

where  $q = \sum_h (n_h/n) F_j^0(y + \rho'_{n;jh})$

$$\frac{dq}{d\rho_{n;jh}} = (n_h/n) f_j^0(y + \rho'_{n;jh}), f_j^0 \text{ the density function of } X_{j\alpha}^0.$$

Since  $\rho'_{n;jh} \xrightarrow{n} 0$  it follows from (i) and (ii), on application of the dominated convergence theorem, that

$$\begin{aligned}(28) \quad \lim_{n \rightarrow \infty} \mu_{ni}^0 &= \int_{-\infty}^{+\infty} \psi'_y \{F(y)\} dF(y) \cdot \lim_{n \rightarrow \infty} \sum_j \sum_h n^{-1} m_{j,h} n_{j,c} \rho_{n;jh} \\ &= \int_{-\infty}^{+\infty} \psi'_y \{F(y)\} dF(y) \cdot \lim_{n \rightarrow \infty} \sum_j n_{j,c} \theta_j / \sqrt{a_n}, \text{ from (10).}\end{aligned}$$

Along the same lines, making use of (11), the right hand side of (28) is also found for

$\lim_{n \rightarrow \infty} \mu_{ni}^*$  which establishes (26) and terminates the proof.

Now define

$$(29) \quad X_n = \begin{bmatrix} \sqrt{n_1} & \sqrt{n_2} & \dots & \sqrt{n_k} \\ \sqrt{n_1} (x_1^{(2)} - \bar{x}^{(2)}) & \sqrt{n_2} (x_2^{(2)} - \bar{x}^{(2)}) & \dots & \sqrt{n_k} (x_k^{(2)} - \bar{x}^{(2)}) \\ \vdots & \vdots & & \vdots \\ \sqrt{n_1} (x_1^{(c)} - \bar{x}^{(c)}) & \sqrt{n_2} (x_2^{(c)} - \bar{x}^{(c)}) & \dots & \sqrt{n_k} (x_k^{(c)} - \bar{x}^{(c)}) \end{bmatrix}$$

$$= (\underline{X}_{n1}, \underline{X}_{n2}, \dots, \underline{X}_{nk})$$

then

$$(30) \quad \underline{X}_n \underline{X}_n' = \begin{bmatrix} n & \underline{0}' \\ \underline{0} & \underline{S}_2 \end{bmatrix}$$

where

$$(31) \quad \underline{S}_2 = \begin{bmatrix} s_{22} & s_{23} & \dots & s_{2c} \\ s_{32} & s_{33} & \dots & s_{3c} \\ \vdots & \vdots & \ddots & \vdots \\ s_{c2} & s_{c3} & \dots & s_{cc} \end{bmatrix}$$

Note that  $S_{11}$  is the determinant corresponding to  $\underline{S}_2$ .

$$(32) \quad \hat{\underline{T}}_n' = (\hat{T}_{n1}, \hat{T}_{n2}, \dots, \hat{T}_{nk})$$

$$(33) \quad \underline{T}_n^0 = (T_{n1}^0, T_{n2}^0, \dots, T_{nk}^0)$$

and

$$(34) \quad \begin{aligned} \Sigma_n &= \{\sigma_{n;ij}\} \\ &= \underline{I}_k - \underline{X}_n' (\underline{X}_n \underline{X}_n')^{-1} \underline{X}_n \end{aligned}$$

where  $\underline{I}_k$  denotes the identity matrix of order  $k$ .

### Theorem 1

Under the conditions of Lemma 1 the asymptotic distribution of  $\hat{\underline{T}}_n$  is normal under  $H_0$  and  $H_n$ , with vector of means  $\underline{0}$  and  $\lim_{n \rightarrow \infty} \nu$  under  $H_0$  and  $H_n$  respectively, and covariance matrix given by  $\lim_{n \rightarrow \infty} (A_{\psi}^2 \Sigma_n)$  where

$$\begin{aligned}
 (35) \quad \nu'_{-n} &= (\nu_{n1}, \nu_{n2}, \dots, \nu_{nk}) \quad \text{and} \\
 \nu_{ni} &= \int_{-\infty}^{+\infty} \psi'_y \{F(y)\} dF(y) \sum_j n_{ij} c_{ij} \theta_j / \sqrt{a_n}.
 \end{aligned}$$

*Proof*

From Lemma 1 it follows that the proof will be completed if the theorem is proved for  $T_{-n}^0$ .

The expressions of the means of the  $T_{ni}^0$  follow by an application of (4.27) of Hajek (1968) i.e.

$$\begin{aligned}
 (E T_{ni}^0 - \mu_{ni}^0)^2 &= O(1)n^{-1} \sum_j n_{ij} c_{ij}^2 \\
 &\xrightarrow{n} 0 \quad (\text{since } \sum_j n_{ij} c_{ij}^2 \text{ is bounded})
 \end{aligned}$$

and from (28).

Subsequently the asymptotic covariance matrix (under  $H_0$  and  $H_n$ ) is determined. From Hajek (1968) (4.26) it follows that

$$(T_{ni}^0 - E T_{ni}^0) - \sum_j \sum_\alpha Z_{ni}^0 (X_{j\alpha}^0) \xrightarrow{n} 0 \text{ in probability.}$$

Along the same lines that we proved (25), we see that  $\sum_j \sum_\alpha Z_{ni}^0 (X_{j\alpha}^0)$  is asymptotically equivalent to  $W_{ni} = \sum_j \sum_\alpha c_{ij} \psi \{F_j^0(X_{j\alpha}^0)\}$ .

Consequently, under  $H_0$  and  $H_n$ ,

$$\begin{aligned}
 &\lim_{n \rightarrow \infty} \text{cov}(T_{ni}^0, T_{nj}^0) \\
 &= \lim_{n \rightarrow \infty} \text{cov}(W_{ni}, W_{nj}) \\
 &= \lim_{n \rightarrow \infty} \sum_h \sum_\alpha \sum_{ih} \sum_{jh} c_{ih} c_{jh} \text{cov}[\psi \{F_h^0(X_{h\alpha}^0)\}; \psi \{F_h^0(X_{h\alpha'}^0)\}] \\
 &= \lim_{n \rightarrow \infty} \sum_h \sum_\alpha c_{ih} c_{jh} \text{var}[\psi \{F_h^0(X_{h\alpha}^0)\}] \\
 &= A_\psi^2 \lim_{n \rightarrow \infty} [\delta_{ij} - \{\sqrt{n_{ij}}/n + \sum_r \sum_t \sqrt{n_{ir} n_{jt}} (x_i^{(r)} - \bar{x}^{(r)})(x_j^{(t)} - \bar{x}^{(t)}) \cdot s_{rt}^{(2)}\}]
 \end{aligned}$$

$$\begin{aligned}
& \text{from (12) and (17) where } \{s_{rt}^{(2)}\} = S_2^{-1} \\
& = A_{\psi}^2 \lim_{n \rightarrow \infty} \{\delta_{ij} - X'_{ni} (X_n X_n')^{-1} X_{nj}\} \\
& = A_{\psi}^2 \lim_{n \rightarrow \infty} \sigma_{n; ij}, \text{ from (34)}
\end{aligned}$$

which establishes the second order moments of  $T_n^0$ .

The proof will be complete if it can be shown that  $\sum_i a_i T_{ni}^0$  is under  $H_0$  and  $H_n$  asymptotically normally distributed for arbitrary constants  $a_i$ . This immediately follows from Theorem 2.2 of Hajek (1968) and the following expression:

$$\sum_i a_i T_{ni}^0 = \sum_i \sum_j \sum_{\alpha} a_i c_{ij} \psi \{R_{j\alpha}^0 / (n+1)\}$$

since the constants  $\sum_i a_i c_{ij}$  satisfy the Noether condition (2.11) of Hajek (1968).

#### 4. Asymptotic distribution of $\hat{Q}_n$ under $H_0$ and $H_n$ . Asymptotic relative efficiency (a.r.e.)

In this section we shall firstly find the asymptotic distribution of  $\hat{Q}_n$  under  $H_0$  and  $H_n$ . For this we require the following well-known lemma:

##### Lemma 2

Let  $X$  be a  $c \times k$  matrix of constants such that  $XX'$  is non-singular. If the  $k$ -variate variable  $Z$  is  $n(\underline{\mu}; I - X'(XX')^{-1}X)$  distributed, then  $Z'Z$  possesses a non-central  $\chi^2(k-c)$  distribution with non-centrality parameter  $\underline{\mu}'(I - X'(XX')^{-1}X)\underline{\mu}$ .

From Theorem 1 and Lemma 2 the following main theorem is derived:

##### Theorem 2

Under  $H_0$  and  $H_n$  the statistic  $\hat{Q}_n$  asymptotically possesses a central and non-central  $\chi^2(k-c)$  distribution respectively with non-centrality parameter

$$(36) \quad \Delta_{\hat{Q}_n} = \lim_{n \rightarrow \infty} \sum_i \sum_j \nu_{ni} \nu_{nj} \sigma_{n; ij} / A_{\psi}^2.$$

Next, we investigate the a.r.e. properties of  $\hat{Q}_n$ . The least squares test statistic for  $H_0$  in the parametric case is given by

$$(37) \quad F_n = \{(n-k)/(k-c)\} \sum_i n_i (\bar{X}_i^{(1)} - \tilde{X}_i)^2 / \{\sum_i \sum_{\alpha} (X_{i\alpha}^{(1)} - \bar{X}_i^{(1)})^2\}$$

where  $\tilde{X}_i = B_{n1} + \sum_r B_{nr} x_i^{(r)}$  and  $B_{nr}$  ( $r = 1, 2, \dots, c$ ) the least squares estimators of the  $\beta_r$ . Under  $H_0$  and the usual normality condition  $F_n$  possesses a central  $F(k - c; n - k)$  distribution. Furthermore, analogously to the derivation above, it can be shown that under  $H_n$  the statistic  $(k - c)F_n$  asymptotically has a non-central  $\chi^2(k - c)$  distribution with non-centrality parameter

$$(38) \quad \Delta_{F_n} = \lim_{n \rightarrow \infty} \sum_i \sum_j \nu_{ni} \nu_{nj} \sigma_n / [\sigma \int_{-\infty}^{+\infty} \psi'_y \{F(y)\} dF(y)]^2, \text{ with } \sigma^2 \text{ defined in section 2.}$$

Therefore, the a.r.e. of the test based on  $\hat{Q}_n$  relative to the one based on  $F_n$  is given by

$$(39) \quad \begin{aligned} e_{\hat{Q}_n, F_n} &= \Delta_{\hat{Q}_n} / \Delta_{F_n} \\ &= \sigma^2 [\int_{-\infty}^{+\infty} \psi'_y \{F(y)\} dF(y)] / A_{\psi}^2, \end{aligned}$$

the usual expression for the a.r.e. of rank tests relative to classical tests.

## 5. Practical application

The tests based on  $\hat{Q}_n$  are only asymptotically distribution-free, a shortcoming which is usually encountered with rank tests for regression problems (see for example Sen (1969) and Adichie (1974)). Through simulation studies it was found that such tests usually are conservative for very small samples (Smit (1979)) and it can be expected that this property will still hold for samples of moderate size. In the construction of the test statistic, use was made of consistent estimators of the regression constants (see (20)). In practice the well-known least squares estimators can be used as well as analogous estimators based on ranks (see for example Adichie (1967)). Smit (1979) found in the case of small samples that the latter type of estimator has a less stable performance than the former and that the use of these rank type estimators tends to have a suppressing effect on the empirical power of the tests. This may lead one to rather use the tests based on the least squares estimators.

Finally, when the sample size is small, the constant  $A_{\psi}^2$ , defined in (17), can be replaced by the commonly used constant

$$(40) \quad A_{\psi, n}^2 = (n-1)^{-1} \sum_{\alpha=1}^n [\psi \{\alpha/(n+1)\} - n^{-1} \sum_{\alpha=1}^n \psi \{\alpha/(n+1)\}]^2$$

which usually has a positive effect on the empirical power of such tests.

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